

Streamlining Epidemiological Model Generation Using Large Language Models

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Abstract—Epidemiological modeling is a critical step in understanding and studying diseases. While conventional modeling techniques, like compartmental or agentic models, exist, supporting tools are not necessarily standardized. Consequently, learning a new tool may incur significant effort. In this work, we explore the use of Large Language Models (LLMs) as a means to quickly transition from requirements, in the form of graphical model representations and initial parameters to a software-aided modeling platform with simulation and verification capabilities. Our methodology employs a metamodel-based language specification for prompt engineering, combined with multimodal inputs comprising textual descriptions and epidemiological diagrams. We evaluate LLM performance across three epidemiological models of varying complexity: a simple SIR compartmental model, a COVID-19 pandemic model, and an HIV transmission model. Initial results demonstrate the potential of LLMs to significantly reduce manual modeling effort while maintaining accuracy in structural representation and parameter extraction. The study provides insights into optimal input configurations and identifies key challenges in automated epidemiological model generation.

Index Terms—epidemiological modeling, large language models, model-driven engineering, automated code generation, SEIR models, prompt engineering

I. PROBLEM STATEMENT AND MOTIVATION

Epidemiological modeling is critical for understanding disease transmission and informing public health policy [1]. Traditional manual construction of differential equation systems is error-prone, requires specialized expertise, and creates significant barriers. The COVID-19 pandemic highlighted critical challenges including transparency, reproducibility, and rapid model development needs [2]. Current limitations include: (1) copy-paste errors in MATLAB and R workflows, (2) lack of tool standardization creating steep learning curves, (3) limited accessibility for non-programmers, and (4) time-intensive development hindering emergency response.

Large Language Models (LLMs) offer opportunities for automating complex technical tasks [3], yet their application to structured scientific modeling, particularly epidemiology, remains unexplored.

II. METHODOLOGY

A. Research Approach

This research investigates LLM capabilities to automatically generate structured epidemiological models from natural language and visual diagrams, focusing on the EpiMDE

(Epidemiological Model-Driven Engineering) platform [4] and its `seirmodel` XML format. The platform bridges traditional epidemiological modeling with model-driven engineering, supporting SEIR (Susceptible-Exposed-Infectious-Recovered) paradigms with complex stratifications.

B. Two-Stage LLM Pipeline

Our pipeline uses Google's Gemini 2.5 Pro [5]: **Stage 1** generates structurally correct `.seirmodel` XML with placeholder values from Main Prompt, Language Specification, and User Input. **Stage 2** calculates final numerical rates using “reason-before-answer” strategy. Figure 1 illustrates a typical SIR model diagram used as input.

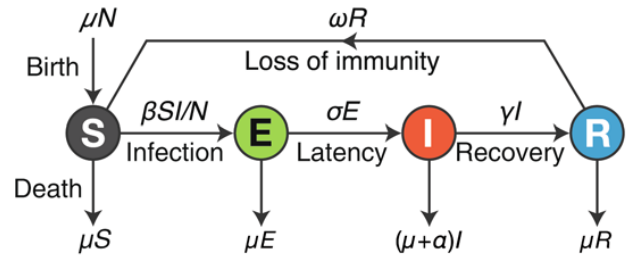


Fig. 1. Example SIR model diagram showing compartments and flows used as multimodal input to the LLM pipeline.

C. Prompt Engineering and Input Design

Three components guide the pipeline: (1) **Main Prompt** - assigns modeling task and XML generation objectives, (2) **Language Specification** - provides metamodel-based format instructions, and (3) **User Input** - contains model-specific data. Our multimodal approach combines epidemiological diagrams (topology) with structured tables (compartments, flows, parameters).

III. EVALUATION AND RESULTS

A. Experimental Design

We evaluated three models: (1) Simple SIR (Susceptible-Infectious-Recovered) for baseline, (2) COVID-19 (22 compartments) [6], and (3) HIV transmission [7]. Each experiment ran five times, showing consistent trends and stable metrics across repetitions, validating methodology robustness. We

compare LLM-generated XML against manual benchmarks using True Positives (TP), False Positives (FP), False Negatives (FN), precision, recall, and F1-score.

B. Results

TABLE I
ELEMENT-LEVEL EVALUATION METRICS

Model	TP	FP	FN	Precision	F1-Score
Simple SIR	7	0	6	1.000	0.700
COVID-19	19	3	14	0.864	0.692
HIV	15	0	17	1.000	0.638

Most False Negatives reflect intelligent constraint adherence rather than failures. In Simple SIR, the LLM correctly omitted births/deaths (unsupported by EpiMDE). In HIV, contact-based flows requiring epidemiological contact matrices were excluded as EpiMDE lacks this support. Perfect precision (1.000) indicates no hallucinations. COVID-19's 14 FNs arose from conservative handling of ambiguous age-stratified flows: with 5 age groups requiring 25 potential inter-group transmissions, the LLM avoided assumptions about symmetric versus asymmetric transmission patterns. However, 3 FPs resulted from cascading errors where missing compartments confused related flows.

IV. KEY FINDINGS AND CONTRIBUTIONS

Key breakthroughs: (1) metamodel-based prompting outperformed templates, (2) multimodal inputs (diagrams + tables) proved essential, (3) two-stage pipeline improved accuracy, (4) LLMs demonstrated intelligent platform constraint reasoning. Contributions include: evaluation framework, prompt engineering methodology, multimodal integration strategies, and constraint management approaches for scientific modeling.

A. Performance Analysis and Limitations

High precision (0.864-1.000) indicates minimal hallucination and strong format adherence. However, significant false negative rates (46.2% for Simple SIR, 42.4% for COVID-19, 53.1% for HIV) warrant critical attention in medical contexts. While many FNs reflect intelligent omission of unsupported features, incomplete generation requires expert review. Missing transmission pathways could underestimate disease spread, while missing compartments could obscure critical population dynamics, potentially resulting in policies failing to protect vulnerable populations. LLM-generated models require thorough expert validation, positioning this as assistive technology for initial construction rather than autonomous generation.

V. IMPLICATIONS AND FUTURE DIRECTIONS

This demonstrates potential for LLM-assisted modeling to: (1) accelerate initial construction, (2) support rapid prototyping with expert validation, (3) reduce implementation errors, and (4) improve reproducibility. However, false negative rates underscore that tools complement rather than replace expert judgment. Frameworks extend to environmental, economic,

and engineering domains requiring structured output generation.

Future work includes: (1) specialized computational engines integration, (2) real-time validation systems, (3) human-in-the-loop systems, (4) multi-platform extension, and (5) domain-specific fine-tuning.

VI. CONCLUSION

This study presents the first comprehensive evaluation of LLMs for automated epidemiological model generation, demonstrating significant potential with important limitations. LLMs can successfully translate natural language descriptions and visual diagrams into structured XML with high precision using appropriate prompt engineering techniques. Key contributions include evaluation frameworks, metamodel-based prompt engineering methodology, multimodal integration strategies, and constraint management approaches applicable to broader scientific modeling contexts.

The findings indicate potential for reducing manual modeling effort in initial development phases, though the substantial false negative rates necessitate expert validation before deployment. Successful generation of complex models ranging from 3 to 22 compartments demonstrates practical applicability as an assistive technology. As LLM technology advances and validation frameworks mature, automated model generation will become increasingly valuable for accelerating scientific research and emergency response capabilities in public health, while maintaining necessary expert oversight for safety-critical applications.

REFERENCES

- [1] F. Brauer, "Compartmental models in epidemiology," in *Mathematical Epidemiology*, F. Brauer, P. van den Driessche, and J. Wu, Eds. Berlin, Heidelberg: Springer, 2008, pp. 19–79.
- [2] J. Sills, C. M. Barton, M. Alberti, D. Ames, J.-A. Atkinson, J. Bales, E. Burke, M. Chen, S. Y. Diallo, D. J. D. Earn, B. Fath, Z. Feng, C. Gibbons, R. Hammond, J. Heffernan, H. Houser, P. S. Hovmand, B. Kopainsky, P. L. Mabry, C. Mair, P. Meier, R. Niles, B. Nosek, N. Osgood, S. Pierce, J. G. Polhill, L. Prosser, E. Robinson, C. Rosenzweig, S. Sankaran, K. Stange, and G. Tucker, "Call for transparency of covid-19 models," *Science*, vol. 368, no. 6490, pp. 482–483, 2020.
- [3] Google, "Gemini api documentation: Gemini 2.5 pro," <https://ai.google.dev/gemini-api/docs/models>, 2024.
- [4] B. Curzi-Laliberté, M. Fokaefs, M. Famelis, and M. Hamdaqa, "EpiMDE: A-model driven engineering platform for epidemiological modeling," ser. MODELS '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 226–236.
- [5] Google, "Image understanding — gemini api documentation," <https://ai.google.dev/gemini-api/docs/image-understanding>, 2025, accessed: 2025-07-13.
- [6] A. R. Tuite, D. N. Fisman, and A. L. Greer, "Mathematical modelling of covid-19 transmission and mitigation strategies in the population of ontario, canada," *CMAJ*, vol. 192, no. 19, pp. E497–E505, 2020.
- [7] C. C. Espitia, M. A. Botina, M. A. Solarte, I. Hernandez, R. A. Riascos, and J. F. C. A. Meyer, "Mathematical model of hiv/aids considering sexual preferences under antiretroviral therapy, a case study in san juan de pasto, colombia," *Journal of Computational Biology*, vol. 29, no. 5, pp. 483–493, 2022. [Online]. Available: <https://www.liebertpub.com/doi/10.1089/cmb.2021.0323>